

## Calculus I Lesson 14

1)  $f(x) = \sqrt{x}$

a)  $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{1cm}} ?$

b)  $f'(0) = \underline{\hspace{1cm}} ?$

c) Is  $f(x)$  differentiable at  $x = 0$ ?

2)  $f(x) = \sqrt{2x+1}$

a)  $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

b)  $f'(-0.5) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

c) Is the function  $f(x) = \sqrt{2x+1}$  differentiable at  $x = -0.5$ ?

$$3) \frac{1}{x-3} + \frac{1}{x-4} = \underline{\quad ? \quad}$$

$$\text{Hint: } \frac{a}{b} + \frac{c}{d} = \frac{a \cdot d + b \cdot c}{b \cdot d} \quad \text{and} \quad \frac{a}{b} - \frac{c}{d} = \frac{a \cdot d - b \cdot c}{b \cdot d}$$

$$4) \frac{1}{x-3} - \frac{1}{x-4} = \underline{\quad ? \quad}$$

$$5) \frac{1}{x+h-3} - \frac{1}{x-3} = \underline{\quad ? \quad}$$

$$6) \frac{1}{x+h-7} - \frac{1}{x-7} = \underline{\hspace{2cm}}$$

$$7) \frac{1}{x+\Delta x-4} - \frac{1}{x-4} = \underline{\hspace{2cm}}$$

8)  $f(x) = \frac{1}{x-4}$

Hint:  $f(x + \Delta x) = \frac{1}{x + \Delta x - 4}$

a)  $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$

b)  $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$

c)  $f'(4) = \underline{\hspace{2cm}}?$

d) Is the function  $f(x) = \frac{1}{x-4}$  differentiable at  $x = 4$ ?

9)  $f(x) = \frac{1}{2x+3}$

Hint:  $f(x + \Delta x) = \frac{1}{2(x + \Delta x) + 3} = \frac{1}{2x + 2\Delta x + 3}$

a)  $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$

b)  $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$

c)  $f'(-1.5) = \underline{\hspace{2cm}}?$

d) Is the function  $f(x) = \frac{1}{2x+3}$  differentiable at  $x = -1.5$ ?

10)  $f(x) = \frac{1}{4x-2}$

Hint:  $f(x + \Delta x) = \frac{1}{4(x + \Delta x) - 2} = \frac{1}{4x + 4\Delta x - 2}$

a)  $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

b)  $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

c)  $f'(0.5) = \underline{\hspace{2cm}} ?$

d) Is the function  $f(x) = \frac{1}{4x-2}$  differentiable at  $x = 0.5$ ?

